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The differential consequences of correcting misinformation for high and low credibility sources

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Communicating accurately is essential for maintaining credibility as a source of information. However, it is unclear whether being inaccurate impacts perceptions of sources equivalently. Across two experiments, we investigated whether correcting inaccuracies differentially impacted credibility for high and low credibility sources (i.e., doctors and politicians, respectively). Participants read a vignette about a high or low credibility source and rated their credibility. Participants were randomly allocated to a correction or a no correction control condition (Experiment 1) or a correction condition with different misinformation frames (highlighting that the source was unintentionally inaccurate, intentionally inaccurate, or uncertain about the information's veracity; Experiment 2). Finally, participants rated belief in the source's misinformation and re-rated credibility. In both experiments, we found that doctors were perceived as more credible and believed more than politicians pre-correction. In Experiment 1, correcting inaccuracies reduced doctors' credibility more than politicians', but belief in misinformation reduced equivalently. In Experiment 2, sources who were uncertain were perceived as less credible pre-correction, and being intentionally inaccurate was associated with the greatest post-correction credibility reduction. However, unlike Experiment 1, corrections impacted the credibility of sources equivalently. These findings underscore the importance of communicating accurately for maintaining trust and demonstrate that the reputational consequences of inaccuracy are sensitive to framing context.

Sources differ widely in their perceived credibility. As such, misinformation attributed to high credibility sources may be perceived very differently than that attributed to low credibility sources. We use the term *misinformation* as an umbrella term for any information contrary to clear scientific evidence (or information that lacks scientific evidence), and *disinformation* for specific cases where the false information has been spread intentionally. One well-established strategy for combating both mis- and disinformation is to correct or "debunk" it (see Prike & Ecker[1] for a review). Correcting false information not only reduces false beliefs, but can also impact the perceived credibility of the sources who spread it. However, studies examining the impact of corrections on source credibility have produced mixed results, with some reporting large effects[2,3] and others finding little-to-no effect[4,5]. One factor that might underlie these divergent results is how credible a source is perceived to be prior to being corrected. The current study seeks to compare the impact of correcting false claims attributed to high and low credibility sources.

Corrections and source credibility

Source credibility is thought to be composed of both trustworthiness (a source's willingness to provide accurate information) and expertise (the extent to which a source is qualified to provide information)[6,7]. Highly credible sources may be penalized more for being inaccurate than less credible sources because people expect the former to be more accurate and honest than the latter[8,9]. In fact, research has found that people have more negative attitudes of politicians from their own party who were inaccurate compared to politicians from an opposing party[10,11] (but, see Galak & Critcher[12]. Evidence also suggests that violating participants' expectations of honesty can lead to a greater reduction in perceived trustworthiness when sources are trusted more at baseline[13]. These findings suggest that sources who are perceived as more credible may have more to lose when they are revealed to be inaccurate. However, to date, no studies have directly compared the impact of correcting misinformation attributed to high and low credibility sources.

Two sources that are ideal for operationalizing high and low credibility are doctors and politicians. In the U.S., these sources consistently fall at opposite ends of the spectrum in terms of public trust[14]. According to the

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Ipsos Trustworthiness Index, a survey that tracks perceptions of trustworthiness across various professions in 32 countries, this is also true internationally[15]. Since the launch of the index in 2018, medical doctors have been consistently perceived as the most trustworthy profession, while politicians have been perceived as the least trustworthy profession[15]. Previous studies examining the impact of correcting inaccuracies attributed to these two sources have shown that corrections typically reduce perceived credibility for doctors, but not always for politicians[4,5,16]. The apparent asymmetrical impact of correcting misinformation for these two sources might be explained by differences in pre-correction expectations of honesty and accuracy. However, as no studies directly compare the impact of corrections across these source types, the driver of this asymmetry requires direct investigation.

Misinformation frame type

The perceived intentions of the misinformation source are also important to consider when investigating why corrections could impact sources differently. For instance, corrections may have a larger impact when it is revealed that the source was deliberately inaccurate (i.e., spreading disinformation) compared to accidentally inaccurate (i.e., made an error). Indeed, sources who are deliberately inaccurate are perceived as less trustworthy than sources who are unintentionally inaccurate[17,18]. Similarly, a study by Ecker et al.[19] found that corrections were more effective at reducing belief in misinformation when participants were told that the source had a conflict of interest compared to when they only saw a correction. Considering that doctors are generally expected to be more honest and competent than politicians[14,15,20], it is possible that doctors may be judged more harshly than politicians when they are deliberately inaccurate.

The degree of certainty with which a source initially conveys misinformation may also influence their credibility. On one hand, expressing uncertainty at the initial presentation of misinformation may serve as a reputational buffer when the source is revealed to be inaccurate[21]. On the other hand, communicating uncertainty at the outset may be reputationally costly enough to outweigh any such benefits. The latter is consistent with the confidence heuristic[22], which suggests that claims made with less certainty are less persuasive because individuals assume that lower certainty implies lower levels of legitimate expertise. Indeed, a study by Lohre et al.[23] found that participants reported feeling more favorable about high certainty sources compared to low certainty sources, regardless of the source's actual expertise. However, this study was conducted with "experts" on the relatively benign and subjective topic of restaurant quality, and it remains unclear whether the same pattern of results would emerge if the topic were more consequential (e.g., policies about cancer).

The current study

The goal of the current study was to assess the impact of correcting false claims attributed to either high or low credibility sources (i.e., medical doctors and politicians). A key question was whether high credibility sources incurred a larger penalty for being inaccurate compared to low credibility sources. Specifically, we assessed the impacts of corrections on source credibility and misinformation belief. We also measured endorsement of the source as an indicator of participants' behavioral intentions to support the source (i.e., intent to visit the doctor's practice or donate to the politician's campaign). In Experiment 1, we attributed four false claims to either a fictitious doctor or politician and then subsequently corrected them. We were also interested in in-group/out-group political differences, given that perceptions of credibility for out-group sources is likely to be lower than for in-group sources[24]. In Experiment 2, we replicated and extended Experiment 1 by assessing the impact of highlighting that the sources were inaccurate either accidentally or intentionally. To assess whether communicating uncertainty protected against post-correction reputation damage, we also included a condition where the source communicated the initial claims with a high degree of uncertainty.

Experiment 1

Methods

The methods and experimental protocol for both experiments were approved by the Northeastern University IRB (#20-12-16) and performed in accordance with all relevant guidelines and regulations, including those in the Declaration of Helsinki.

Design

This study followed a 2 × 2 × 2 within-between design with within-subjects factor timepoint (pre-correction vs. post-correction) and between-subjects factors correction (no-correction control vs. correction) and source (high credibility vs. low credibility). To reflect the most and least trusted professions around the globe[15], we used a doctor as the high credibility source and a politician as the low credibility source. Participants were randomly allocated to the high credibility (doctor) or low credibility (politician) source condition. In the political source condition, we counterbalanced source partisanship such that participants were exposed to either a Democratic or Republican source. This resulted in approximately half of those in the political source condition being exposed to a political in-group source and half being exposed to a political out-group source.

Participants

Participants were a representative sample of 744 U.S. adults (based on sex, age, ethnicity, and political affiliation) over the age of 18 from Prolific. Our preregistered exclusion criteria excluded participants who completed the task too quickly (i.e., outliers with regards to completion time; $n = 0$) and participants who indicated that their data should be excluded due to a lack of effort ($n = 1$). We also preregistered that we would exclude political independents to analyze political in-group vs. out-group differences ($n = 95$). Our final sample included 648 people, with 311 males, 326 females, and 11 individuals who were non-binary, preferred to self-describe, or who chose not to disclose their gender (Age Range = 18–85 years, M_{age} = 45.71, SD_{age} = 15.46). See Supplementary

Table 1 for full demographic information across experimental conditions. Participants were paid $2 for a 10-minute task.

Sample size justification

Effect sizes associated with correcting misinformation attributed to a credible health source are generally large when compared to a no-correction control (e.g., $d = 1.18$ for belief in misinformation[2]. However, prior research comparing a correction's impact across misinformation sources is limited. As such, we used the effect size between an in-group and an out-group politician, which appears to be small to moderate ($\eta_p^2 = 0.04 - .23$[3,5]). Our primary analysis for misinformation belief and source credibility was an ANCOVA with factors source (doctor vs. politician) and correction (correction vs. control), controlling for pre-correction ratings. Using the smallest known effect size, G*Power3 recommended a minimum sample size of 314 participants this analysis (with $\alpha = 0.05$, power = 0.95; $\eta_p^2 = 0.04$; $f = .20$[3], for an ANCOVA fixed effects, main effects, and interactions[25]. We boosted our N to 628 so that our analysis focusing on participants allocated to the political conditions (in-group, out-group) was sufficiently powered (an ANOVA with factors source [in-group vs. out-group] and partisanship [Democrat vs. Republican]). Finally, we further increased the initial sample to 736 to facilitate the exclusion of political independents (approximately 15% of Prolific sample[26]).

Procedure and stimuli

Participants provided informed consent to participate and then were introduced to the sources. Participants in the high credibility source condition were introduced to a fictional doctor (Dr. Alex Stephens), while those in the low credibility source condition were introduced to a fictional Senator (Senator Alex Stephens). Participants were told "Please think of Dr./Senator Stephens as you would your typical medical doctor/politician." Senator Stephen's partisanship was counterbalanced so that for half of the participants the Senator was a Republican, and for the other half the Senator was a Democrat.

Next, all participants read a brief vignette describing an interview with their target source on a fictional piece of legislation called the "Cancer Prevention and Protection Bill", which proposed implementing tax increases/cuts on products ostensibly linked to cancer. We used a health-relevant bill so that both politicians and doctors could reasonably hold a high level of relevant expertise. Four false claims about cancer were embedded in these vignettes, all attributed to the target source (e.g., "Senator Stephens confirmed that artificial sweeteners can increase the risk of developing cancer, and supported higher taxes…"; see Supplementary Methods for full vignettes). Claims were selected from Swire-Thompson et al.[2] based on having sufficiently high believability. Vignettes were identical except for the source attribution.

Participants subsequently rated how credible they found the source to be, their belief in the four false claims, and their support for the Cancer Prevention and Protection Bill. Belief in misinformation was measured on 1–7 Likert scales ("How much do you believe the following to be true?"; 1 = *not at all*, 7 = *very much so*). Source credibility was measured using a 12-item composite (1 = *not at all*, 7 = *extremely*). The composite contained two items measuring general credibility ("In general, how credible do you find [Dr./Senator] Stephens?"; "Do you think that [Dr./Senator] Stephens is a credible source when it comes to the Cancer Prevention and Protection Bill?") and ten items from the trustworthiness and expertise subscales of the Credibility Semantic scale[27]. The Credibility Semantic scale contains two sets of five adjectives measuring both trustworthiness (dependable, honest, reliable, sincere, and trustworthy) and expertise (expert, experienced, knowledgeable, qualified, and skilled). While this scale has not been used to assess the credibility of political sources, this scale has been applied to health sources[2,28] and has previously been found to have excellent reliability (Cronbach's $\alpha = .96$[29]). Support for the fictional Cancer Prevention and Protection Bill was measured with four inference questions related to the misinformation items (e.g., "There should be higher taxes on foods and beverages containing artificial sweeteners due to their cancer-causing properties."), which we combined to form a composite (see Supplementary Results).

Participants were then randomly allocated to either the correction or control condition. In the correction condition, participants received corrections for each false claim presented in the initial vignette. Each correction included an explanation referencing a credible source (e.g., National Cancer Institute; see Supplementary Methods for all corrections). Participants in the control condition read an unrelated article about the benefits of slow running, matched to the correction condition on length. Immediately afterwards, all participants re-rated source credibility, their belief in each false claim, and their support for the bill. Participants then rated their endorsement of the source. Endorsement of the doctor was measured by a composite of two items ("If Dr. Stephens were a local doctor, how likely would you be to visit his practice?"; "If Dr. Stephens published a book, how likely would you be to purchase it"; 1 = *not at all*, 7 = *very much so*). Endorsement of the politician was also measured using a two-item composite ("If you lived in Senator Stephens' state, how likely would you be to vote for him?"; "How likely would you be to donate to Senator Stephens' campaign?"). Although these endorsement items may not be directly comparable across sources (i.e., the context of voting and consulting a physician are different) we included these measures to assess participants' attitudes and behavioral intentions in a realistic manner[2]. Finally, participants were debriefed on the aims of the study, and participants in the control condition were provided with corrections for each false claim presented in the study.

Pre-registration

The following hypotheses and analyses were preregistered at https://osf.io/xthsg/, and we conducted the experiment in September, 2024. Survey design, experimental materials, and debriefing sheets can be found in the OSF repository. We hypothesized that prior to corrections, doctors would be rated as more credible and believed more than politicians. We also hypothesized that corrections would reduce credibility, belief, and endorsement relative to control, and that credibility and endorsement would reduce more for doctors than politicians. Based on

prior evidence that corrections typically reduce belief in misinformation regardless of source[1,5,30], we predicted that belief would reduce equally for both sources. For participants exposed to the politician, we predicted that political in-group sources would be perceived as more credible and believed more than political out-group sources pre-correction[24]. However, we also predicted that corrections would reduce belief equivalently for in-group and out-group sources[3,5,30]. Finally, we predicted that pre- and post-correction, political in-group/out-group differences in belief would be equivalent for Democrats and Republicans. See Supplementary Table 2 for whether each hypothesis was supported or rejected.

Results

All analyses were performed in R version 4.4.3[31]. All statistical tests reported in this manuscript are two-tailed, and *p*-values for planned comparisons and post-hoc tests are Tukey corrected. We present results from our pre-registered ANOVA/ANCOVA models in the main text. We reanalyzed our data using linear mixed effects models for robustness given that the assumption of normality was violated for the ANCOVA models. We note in the main text the rare cases where the results of these two analytic approaches diverge (see Supplementary Results for full analyses). We additionally reanalyze all null results using Bayesian analyses and report corresponding Bayes factors (BFs), given that this method can quantify relative evidence favoring the null or alternative hypothesis. The findings can be expressed as either BF_{10}, which quantifies support for the alternative hypothesis, or BF_{01}, which quantifies support for the null hypothesis. A BF between 1 and 3 provides anecdotal evidence, 3–10 provides moderate evidence, 10–30 provides strong evidence, 30–100 provides very strong evidence, and a BF greater than 100 constitutes extreme evidence[32]. We note that our experimental groups were well-balanced across demographic groups, with no significant differences between the experimental conditions in age (p =.562), education (p =.184), gender (p =.247), or partisanship (p =.813).

Perceived source credibility

First, we confirmed that doctors were indeed perceived as more credible than politicians by conducting a *t*-test on source credibility across these two sources, $t(638) = 8.58$, p <.001, 95% CI = [0.74, 1.19], Cohen's d = 0.67. To assess the impact of corrections on perceived source credibility, we conducted an ANCOVA with factors correction (correction vs. control) and source (doctor vs. politician) on post-correction credibility scores, entering pre-correction credibility as a covariate. This produced a main effect of correction, $F(1, 643) = 514.93$, p <.001, $MSE = 1.21$, $\eta_p^2 = 0.45$. Planned comparisons revealed that corrections reduced credibility ratings relative to control for both doctors, $t(643) = 18.46$, p <.001, 95% CI = [1.94, 2.56], Cohen's $d = 2.05$, and politicians, $t(643) = 13.62$, p <.001, 95% CI = [1.35, 1.98], Cohen's $d = 1.52$. However, the main effect was qualified by a significant correction × source interaction, $F(1, 643) = 11.42$, p <.001, $MSE = 1.21$, $\eta_p^2 = 0.02$, suggesting that the impact of corrections on credibility was greater for the doctor compared to the politician (see Fig. 1, Panel A). When focusing exclusively on the post-correction timepoint, doctors and politicians show similar credibility scores (p =.185). However, Bayesian analyses do not provide meaningful evidence either for or against the null hypothesis ($BF_{01} = 1.36$).

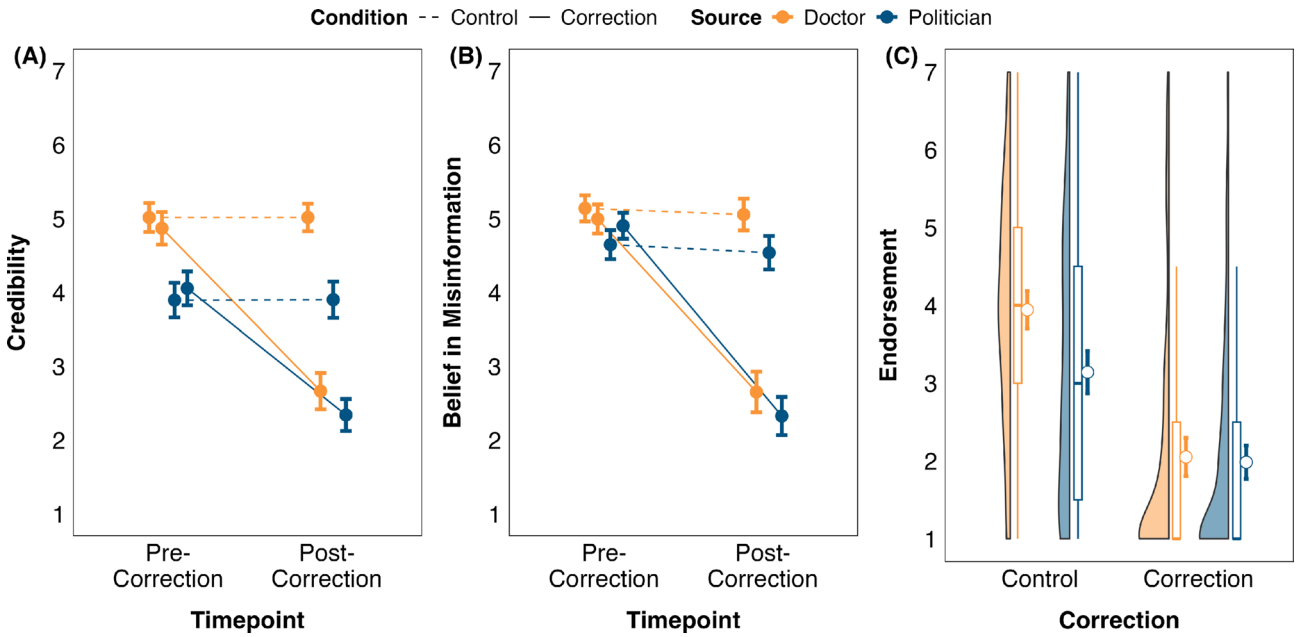


Fig. 1. Point estimates of credibility (A), belief (B), and endorsement (C) in the control (dashed lines) and correction conditions (solid lines). Orange = doctor and blue = politicians. Point estimates reflect means, error bars reflect 95% confidence intervals, and lines in boxplots reflect the median.

Belief in misinformation

Prior to corrections being presented, false claims attributed to the doctor were believed significantly more than those attributed to the politician, $t(646) = 3.02$, $p = .003$, 95% CI = [0.10, 0.48], with a small effect size of $d = 0.24$. To investigate whether correction effects varied by source, we conducted an ANCOVA on misinformation belief with factors correction and source, entering pre-correction belief as a covariate. This produced a significant main effect of the correction, $F(1, 643) = 577.98$, $p < .001$, $MSE = 1.55$, $\eta_p^2 = 0.47$, indicating that corrections reduced belief in misinformation relative to the no-correction control (see Fig. 1, panel B). Consistent with predictions, we did not find a main effect of source, $F(1, 643) = 3.34$, $p = .068$, $BF_{01} = 2.43$, nor a correction × source interaction, $F(1, 643) = 0.50$, $p = .480$, $BF_{01} = 6.87$. Together, this indicates that corrections reduced belief in misinformation for doctors and politicians similarly.

Endorsement

We next examined participants' endorsement of the sources. We conducted a 2 × 2 between-subjects ANOVA on endorsement ratings with factors correction (correction vs. control) and source (doctor vs. politician), which produced a main effect of the correction, $F(1, 644) = 150.29$, $p < .001$, $MSE = 2.50$, $\eta_p^2 = 0.19$, and source, $F(1, 644) = 12.11$, $p < .001$, $MSE = 2.50$, $\eta_p^2 = 0.02$. Planned comparisons demonstrated that endorsement was lower in the correction condition than the control condition for both the doctor and the politician. There was also a correction × source interaction, $F(1, 644) = 8.64$, $p = .003$, $MSE = 2.50$, $\eta_p^2 = 0.01$, driven by the control-correction differences being greater for the doctor, $t(644) = 10.77$, $p < 001$, 95% CI = [1.44, 2.34], $d = 1.72$, than the politician, $t(644) = 6.58$, $p < .001$, 95% CI = [0.71, 1.61], $d = 0.74$; (see Fig. 1, panel C).

In-group/out-group

We next turned our attention to whether our outcomes of interest varied according to whether the politician was a member of participants' in-group or out-group. Consistent with predictions, the political in-group source was perceived as more credible than the political out-group source prior to the correction, $t(317) = 3.32$, $p = .001$, 95% CI = [0.22, 0.87], $d = 0.37$. Post-correction reductions in credibility did not differ across in-group and out-group political sources, or across political parties (all ps ≥ .105, BF_{01}s ≥ 1.69). Turning to misinformation belief, there was no significant difference in belief between misinformation from in-group and out-group politicians pre-correction, $t(306) = 0.86$, $p = .389$, 95% CI = [−0.15, 0.39], $BF_{01} = 5.68$. We also found no differences in belief reduction according to source congruence, $F(1, 318) = 2.64$, $p = .105$, $BF_{01} = 2.77$, or political partisanship, $F(1, 318) = 0.01$, $p = .921$, $BF_{01} = 8.01$. Full post-correction and political partisanship results are available in the Supplementary Results.

Discussion

In Experiment 1, we were interested in whether corrections differentially impacted high credibility (i.e., doctors) and low credibility (i.e., politicians) sources. Consistent with predictions, we found that corrections (a) were associated with larger reductions in credibility for doctors compared to politicians and (b) reduced belief equally for doctors and politicians. We also found that doctors were endorsed more than politicians in the control condition, but that endorsement was lower for doctors than politicians when they were corrected. On the whole, these findings indicate that corrections may be more impactful for high credibility sources than low credibility sources.

These findings provide preliminary insight into the relationship between correcting inaccuracies and source credibility. However, previous research suggests that the intentions of the misinformation source are also important to consider when examining why corrections may impact sources differently. For example, sources who are perceived to be intentionally inaccurate are perceived as less trustworthy than sources who are unintentionally inaccurate[18]. It is plausible that the impact of source credibility and intentions may interact, with the consequences of dishonesty higher for sources who are trusted more to begin with[13]. Revealing that a source is intentionally inaccurate may also lead participants to believe less in claims that source makes in the future[19]. In Experiment 2, we manipulated the intentions of the misinformation source in order to investigate whether high credibility sources are penalized more than low credibility sources for lying (i.e., spreading disinformation). We also examined whether expressing uncertainty during the initial communication of misinformation mitigated reputational damage caused by corrections, and whether this impacted high credibility and low credibility sources differently. We note that our conceptualization of uncertainty is broadly aligned with "deficient uncertainty" (i.e., a source being uncertain of the current state of the scientific literature[33].) In order to simplify our design and maximize initial differences in perceived source credibility, we used out-group politicians only. Finally, we included two additional outcome measures: how much participants' expectations of the source were violated, and how much they believed in new claims attributed to the source. These measures provide further insight into what drives differences in correction efficacy, and allows us to examine whether reduced credibility leads participants to discount future claims made by the sources[19].

Experiment 2

Methods

Design

Experiment 2 followed a 2 × 2 × 3 within-between design with within-subjects factor timepoint (pre-correction vs. post-correction) and between-subjects factors source (doctor vs. out-group politician) and misinformation frame type (unintentional vs. intentional vs. uncertain).

Participants

Participants were a representative sample of 874 U.S. adults (based on sex, age, and political affiliation) over the age of 18 from Prolific. Our preregistered exclusion criteria excluded participants who completed the task too quickly (i.e., outliers with regards to completion time; $n = 0$), participants who indicated that their data should be excluded due to a lack of effort ($n = 0$), and participants who did not identify as a Democrat or Republican ($n = 85$). Our final sample included 789 people, with 380 males, 399 females, and 10 individuals who were non-binary, preferred to self-describe, or who chose not to disclose their gender (Age Range = 18–88 years, M_{age} = 46.01, SD_{age} = 15.55). See Supplementary Table 3 for full demographic information across experimental conditions. Participants were paid $2.40 for a 12-minute task.

Sample size justification

In Experiment 1, the ANCOVA with factors correction condition (control vs. correction) and source (doctor vs. politician) on source credibility ratings produced a significant interaction with a small effect size ($\eta_p^2 = 0.02$, $f = 0.14$). To detect an effect size equally small in Experiment 2 across three misinformation frames (i.e., compare uncertainty vs. accidentally vs. deliberately inaccurate), G*Power3 recommended a sample size of 760 participants (with $\alpha = 0.05$, power = 0.95, for an ANCOVA with fixed effects, main effects and interactions[25]. Because we were interested in differences in our outcomes between doctors and out-group politicians, we excluded political independents from our main analyses (~ 15%[26]). To account for these exclusions, we boosted the initial sample size to 874 participants.

Procedure and stimuli

The procedure and stimuli were similar to those of Experiment 1, with a few key differences. Participants provided informed consent and were again introduced to either a high or low credibility source. However, unlike Experiment 1, low credibility sources were only presented as members of participants' political out-group, such that participants who reported identifying as Democrats viewed only a Republican source and vice versa. After being introduced to the source, participants were randomly allocated to receive an unintentional, intentional, or uncertain misinformation frame. Participants in the unintentional and intentional frame conditions read the "Cancer Prevention and Protection Bill" vignette from Experiment 1. By contrast, the uncertain condition contained the same four false claims but in the context of a source who was unsure about the claim's veracity (e.g., "[Dr./Senator Stephens] said that he didn't know if there was evidence supporting the claim that antiperspirants containing aluminum increase the risk of breast cancer."; see Supplementary Methods for full vignette texts). Participants across all conditions provided the same credibility and belief ratings as Experiment 1.

All participants subsequently received a correction, which varied across the misinformation frame conditions. The unintentional frame included information that the source had made inaccurate statements completely accidentally, whereas the intentional frame revealed that the source had lied to boost sales for a holistic health company they owned. The uncertain frame repeated that the source was unsure about the veracity of the statements they had made (e.g., "Dr. Stephens said that it was too early to tell whether artificial sweeteners increase the risk of developing cancer. This is false…"). All three frames also highlighted that the source issued a press release clarifying their errors and reaffirming their support for evidence-based health policies (see Supplementary Methods for full vignette texts). Next, all participants re-rated source credibility and their belief in each false claim. Participants also rated three items measuring the extent to which the source violated their expectations ("How surprised were you by the way that [Dr./Senator] Stephens communicated?"; "How disappointed were you by the way that [Dr./Senator] Stephens communicated?"; and "Did [Dr./Senator] Stephens act in the way that you expected?", which was reverse scored). These items demonstrated acceptable reliability for a 3-item measure (Cronbach's $\alpha = 0.64$). Participants were also asked the extent to which they believed four new (uncorrected) false cancer claims attributed to the sources. These claims were also taken from Swire-Thompson et al.[2] based on their sufficiently high believability.

Pre-registration

This experiment was conducted in April 2025. The following hypotheses and analyses were preregistered at https://osf.io/xthsg/ and we label all exploratory analyses as such below. We hypothesized that pre-correction (a) doctors would be rated as more credible and believed more than politicians, and (b) sources expressing uncertainty would be perceived as less credible and less believable than sources who were certain[23]. We also predicted that doctors would show larger reductions in perceived credibility and larger expectation violations than politicians post-correction[12,14]. Consistent with Experiment 1, we predicted no differences in belief reduction or belief in new misinformation across source conditions[1,5,30]. Finally, we predicted that the intentional condition would show the largest reductions in credibility and misinformation belief, the strongest expectation violations, and also the lowest belief in new misinformation[17,18]. By contrast, the uncertainty condition was expected to have the smallest reductions in both credibility and belief.

Results

Our experimental conditions were well balanced across demographics, with no significant differences between groups in age ($p = .736$), education ($p = .266$), gender ($p = .742$), or political party ($p = .784$). We present results from our pre-registered ANOVA/ANCOVA models in the main text and include linear mixed model analyses in the Supplementary Results given that the assumption of normality was violated. We note where the results of these two analytic approaches diverge in the main text.

Perceived source credibility

First, we examined pre-correction differences in source credibility for the doctor and the politician. A 2 (source: doctor vs. politician) × 3 (misinformation frame type: unintentional vs. intentional vs. uncertain) between-subjects ANOVA on pre-correction credibility ratings produced a main effect of source, $F(1, 783) = 104.23$, $p < .001$, $MSE = 2.26$, $\eta_p^2 = 0.12$, and misinformation frame, $F(1, 783) = 25.75$, $p < .001$, $MSE = 2.26$, $\eta_p^2 = 0.06$, but no source × misinformation frame type interaction, $F(2, 783) = 0.79$, $p = .456$, $BF_{01} = 18.20$. Replicating Experiment 1, doctors were perceived as more credible than politicians prior to being corrected, $t(783) = 10.21$, $p < .001$, 95% CI = [0.88, 1.30], $d = 0.73$. Additionally, the uncertain frame was associated with lower pre-correction credibility ratings than both the unintentional, $t(783) = 6.21$, $p < .001$, 95% CI = [0.51, 1.12], $d = 0.54$, and intentional frames, $t(783) = 6.21$, $p < .001$, 95% CI = [0.51, 1.12], $d = 0.54$. There was no pre-correction difference between the intentional and unintentional misinformation frame conditions, $p > .999$, $BF_{01} = 10.29$.

To assess how source credibility reduced across the sources and misinformation frames following a correction, we conducted an ANCOVA on source credibility ratings with factors source and misinformation frame, entering pre-correction credibility as a covariate. We did not find a main effect of source, $F(1, 782) = 0.57$, $p = .450$, $BF_{01} = 8.80$, indicating that credibility reduced similarly for doctors and politicians, contrary to our prediction. There was also no source × misinformation frame type interaction, $F(2, 782) = 2.21$, $p = .111$, $BF_{01} = 4.88$. However, there was a main effect of misinformation frame type, $F(2, 782) = 20.76$, $p < .001$, $MSE = 1.63$, $\eta_p^2 = 0.05$. Planned comparisons showed that credibility reduced more in the intentional condition compared to both the unintentional, $t(782) = 4.35$, $p < .001$, 95% CI = [0.22, 0.75], $d = 0.38$ and uncertain misinformation frame types, $t(782) = 6.27$, $p < .001$, 95% CI = [0.45, 0.98], $d = 0.56$ (see Fig. 2). However, there was no difference in credibility reduction between the uncertain and unintentional misinformation frames , $p = .110$, $BF_{01} = 6.92$. At the post-correction timepoint, credibility was lower in the intentional frame than both the unintentional, $t(783) = 3.90$, $p < .001$, 95% CI = [0.19, 0.78], $d = 0.34$, and uncertain frames, $t(783) = 5.26$, $p = .008$, 95% CI = [0.08, 0.66], $d = 0.26$.

Misinformation belief

To examine pre-correction differences in misinformation belief, we conducted a 2 × 3 between-subjects ANOVA with factors source (doctor vs. politician) and misinformation frame type (unintentional vs. intentional vs. uncertain). This produced a main effect of source, $F(1, 783) = 10.94$, $p < .001$, $MSE = 1.56$, $\eta_p^2 = 0.01$, and misinformation frame, $F(1, 783) = 5.93$, $p = .003$, $MSE = 1.56$, $\eta_p^2 = 0.02$, but no source × misinformation frame

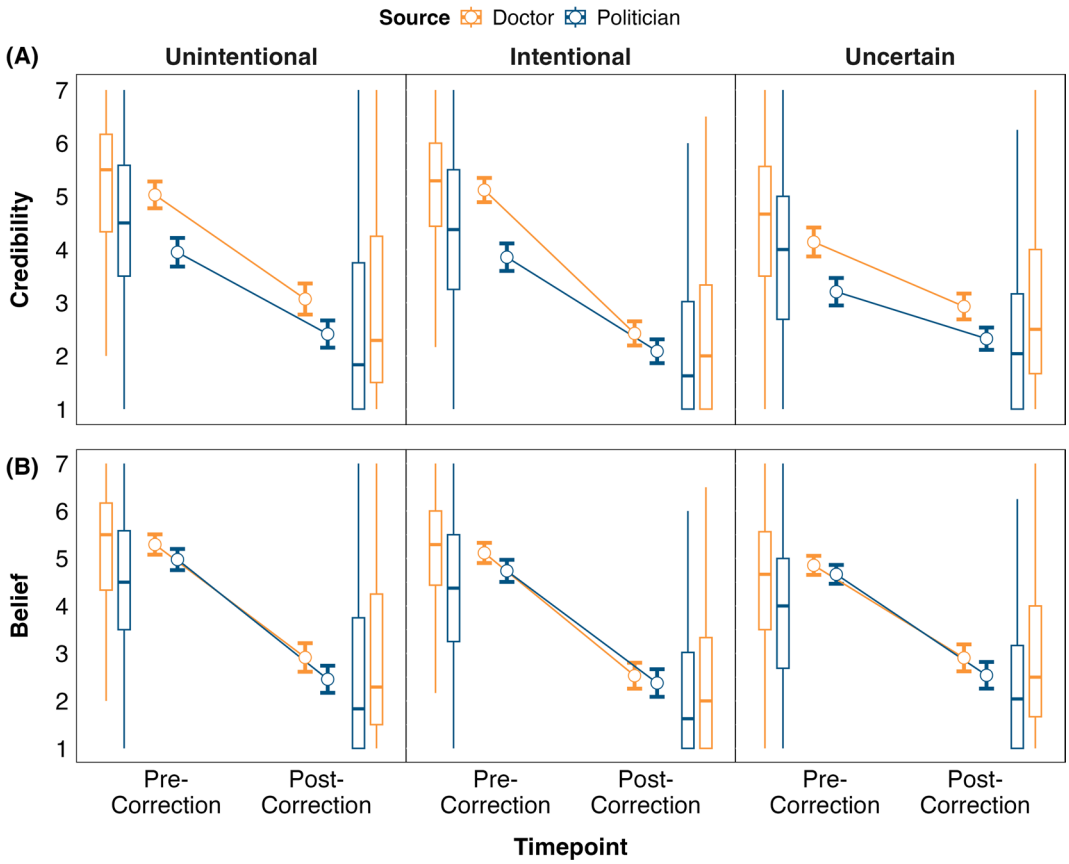


Fig. 2. Boxplots and point estimates for credibility (top) and belief (bottom) reduction by condition. Orange reflects estimates for doctors and blue reflects estimates for politicians. Lines in boxplots indicate the median, point estimates represent the mean, and error bars represent 95% confidence intervals.

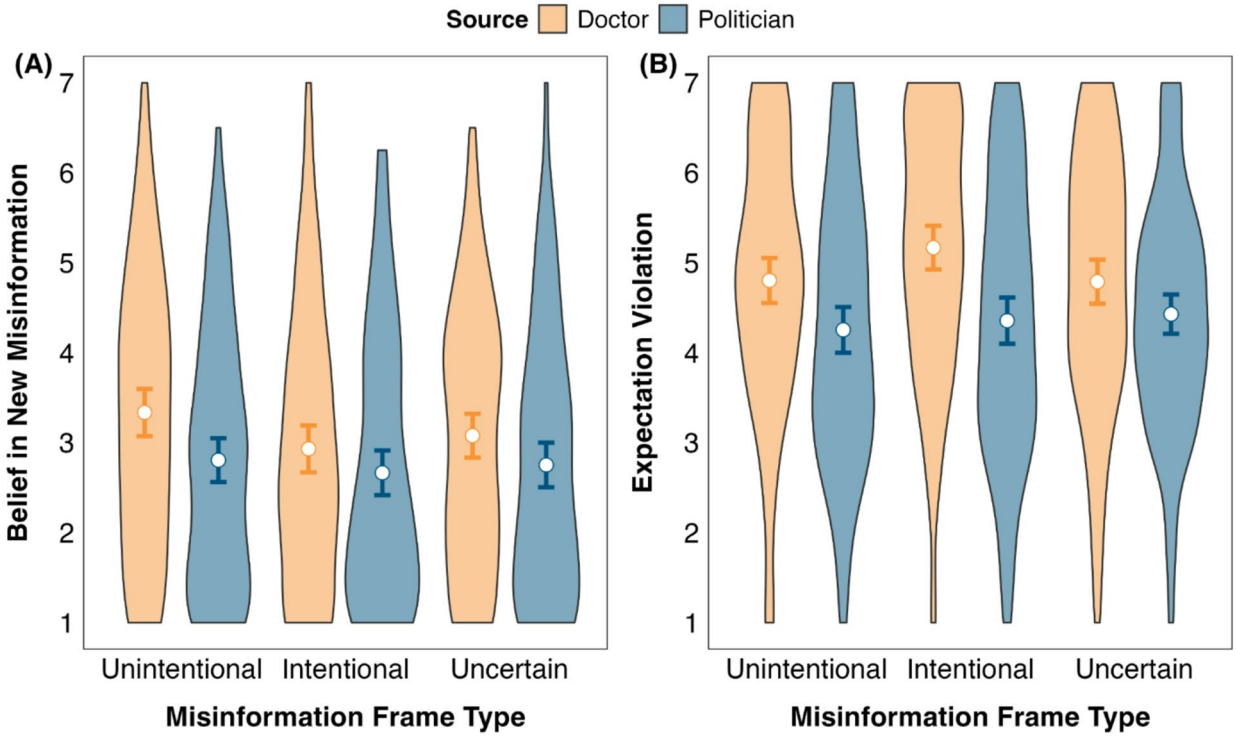


Fig. 3. Distributions and point estimates for belief in new misinformation (A) and expectation violations (B) by condition. Orange = doctors and blue = politicians. Point estimates represent the mean, error bars represent 95% confidence intervals, and violins provide distributional information.

type interaction, $F(2, 783) = 0.39$, $p = .678$, $BF_{01} = 24.70$. As seen in Fig. 2, misinformation attributed to the doctor was believed more than misinformation attributed to the politician pre-correction, $t(783) = 3.31$, $p = .001$, 95% CI = [0.12, 0.47], $d = 0.24$. Regarding misinformation frame, planned comparisons showed that misinformation presented with uncertainty was believed *less* than misinformation in the unintentional frame, $t(783) = 3.44$, $p = .002$, 95% CI = [0.12, 0.63], $d = 0.30$. However, there were no differences between the intentional and unintentional frames, ($p = .139$, $BF_{01} = 1.96$), or the intentional and uncertain frames, ($p = .275$, $BF_{01} = 3.25$).

To examine reductions in misinformation belief, we conducted an ANCOVA with factors source (doctor vs. politician) and misinformation frame type (unintentional vs. intentional vs. uncertain) on misinformation belief, entering pre-correction belief as a covariate. There was no main effect of source, $F(1, 782) = 2.79$, $p = .095$, $BF_{01} = 3.06$, nor a source × misinformation frame type interaction, $F(2, 782) = 0.91$, $p = .403$, $BF_{01} = 14.80$. Yet, we did find a small but statistically significant effect of the misinformation frame, $F(2, 782) = 3.30$; $p = .037$; $MSE = 2.45$, $\eta_p^2 = 0.01$. However, we note that this effect did not cross the threshold of significance in the linear mixed-model analyses ($p = .051$; see Supplementary Results). Planned comparisons for the above ANCOVA showed that misinformation belief reduced more in the intentional condition compared to the uncertain condition, $t(782) = 2.55$, $p = .030$, 95% CI = [0.03, 0.67], $d = 0.22$. However, there were no statistically significant differences in belief reduction between the intentional and unintentional misinformation frames, $p = .593$, $BF_{01} = 3.03$, or unintentional and uncertain misinformation frames, $p = .264$, $BF_{01} = 10.04$. When analyses were isolated to the post-correction timepoint, there was no statistically significant difference in belief across misinformation frames, $ps \geq .136$, $BF_{01} = 10.02$.

To investigate the potential effects of source and misinformation frame on belief in new misinformation, we conducted a between-subjects ANOVA with factors source and misinformation frame. This produced a main effect of source, $F(2, 783) = 12.81$, $p < .001$, $MSE = 2.16$, $\eta_p^2 = 0.02$, with doctors still believed more than politicians post-correction (see Fig. 3, panel A). However, there was again no main effect of misinformation frame, $F(1, 783) = 2.28$, $p = .104$, $BF_{01} = 7.88$, or source × misinformation frame type interaction, $F(2, 783) = 0.58$, $p = .561$, $BF_{01} = 20.76$.

Expectation violations

Finally, we assessed the extent to which participants' expectations of the source were violated after it was revealed that they were inaccurate. A 2 × 3 between-subjects ANOVA with source (doctor vs. politician) and misinformation frame type (unintentional vs. intentional vs. uncertain) on the expectation violations composite revealed a main effect of source, $F(1, 783) = 31.54$, $p < .001$, $MSE = 2.06$, $\eta_p^2 = 0.04$, showing that the doctor violated expectations more than the politician. As seen in Fig. 3 (panel B), there was no main effect of misinformation frame type, $F(2, 783) = 1.80$, $p = .167$, $BF_{01} = 12.72$, or source × misinformation frame type interaction, $F(2,$

783) = 1.65, p =.193, BF_{01} = 7.93. This indicates that a doctor being inaccurate violated expectations more than a politician being inaccurate, however, the reason for being inaccurate did not meaningfully alter this effect.

Discussion

Consistent with predictions, we found that doctors were perceived as more credible than politicians, and that sources expressing uncertainty were perceived as less credible prior to correction. Credibility for both sources reduced equivalently post-correction, and revealing that sources were intentionally inaccurate was associated with the greatest reductions in perceived credibility. In terms of misinformation belief, we found that doctors were believed more than politicians pre-correction, but that corrections reduced belief equivalently. There were no significant differences between the uncertainty and unintentional conditions pre-correction, but there were significant differences between the uncertain and intentional conditions. This was despite the content in the intentional and unintentional conditions being equivalent at this timepoint, suggesting that these effects may not be robust. Moreover, while belief reduction was slightly greater in the intentional relative to the uncertain misinformation frames, all other contrasts were non-significant. Regarding expectation violations, we found that inaccuracies violated participants' expectations of doctors more than their expectations of politicians, but that misinformation frame type did not impact expectations. Finally, despite corrections producing higher expectation violation scores for the doctor, we found that participants still believed new claims attributed to the doctor more than new claims attributed to the politician.

General discussion

The current study aimed to compare correcting misinformation attributed to high credibility (doctors) and low credibility (politicians) sources . In both experiments, the doctor was perceived as more credible and believed more than politician pre-correction, and corrections reduced perceived credibility and misinformation belief for both sources. In Experiment 1, we found that corrections were associated with greater reductions in credibility and lower endorsement ratings for doctors than politicians. Experiment 2 demonstrated that expressing uncertainty led to sources being perceived as less credible. In fact, being uncertain at the outset proved so detrimental to perceptions of credibility that it outweighed any potential protective benefit it may have provided post-correction. We also found that credibility reduced the most when sources were intentionally inaccurate, and that inaccuracies attributed to doctors were associated with larger expectation violations than inaccuracies attributed to politicians. Despite this, new misinformation attributed to the doctor was still believed more than new misinformation attributed to the politician.

Credibility

We found that corrections reduced perceived source credibility for both doctors and politicians, which differs from several previous studies reporting that corrections did not impact politicians' credibility[2,4,5,16]. This divergence may be because participants lacked strong priors about the fictional sources used in these experiments, unlike the real political figures used in past research (e.g., Donald Trump[4,5]). In Experiment 1, we also found that the impact of corrections was more pronounced for doctors. However, there were no source differences in Experiment 2, suggesting that how misinformation is framed may overshadow the impact of who is spreading it. Indeed, being found to be intentionally inaccurate led to the largest credibility loss for both sources[17,18]. The absence of source differences in Experiment 2 is notable given that expectations were violated more for doctors than politicians. Despite participants being more disappointed in the doctor and more surprised by their behavior, they were still rated as more credible than politicians. This result speaks to the persistence of pre-correction credibility differences between these two sources[15].

We also found that sources who expressed uncertainty pre-correction were perceived as less credible, and that this was costly enough to override any benefit that hedging may have had against being explicitly inaccurate post-correction. This might reflect participants interpreting uncertainty as a cue for low expertise[22]. We note that the uncertainty condition used in the current study was more aligned with "deficient uncertainty", a source being uncertain of the current state of the scientific literature, rather than there being uncertainty expressed about agreement in the scientific literature (i.e., "consensus uncertainty") or with the precision of the scientific evidence itself (i.e., "technical uncertainty"[34]). It is plausible that manipulating different types of uncertainty may impact our outcomes differently[34,35]. For instance, prior work has shown that expressing uncertainty about scientific results due to factors like measurement error or statistical assumptions *increases* perceived credibility, whereas uncertainty communicated about scientific consensus has shown the opposite effect[34]. In many ways, the reduction of credibility resulting from our sources' lack of knowledge is a rational and appropriate response. Future research should investigate the conditions under which uncertainty may increase or reduce credibility, particularly for professions where uncertainty can be fundamental for communicating information accurately (e.g., doctors and scientists).

Belief in misinformation

In both experiments, we found that corrections were equally effective at reducing belief in misinformation spread by doctors and politicians. These findings align with prior research showing that corrections are broadly effective at reducing false beliefs irrespective of source[1,5,30,36]. We also found that corrections reduced belief similarly for sources who were intentionally compared to unintentionally inaccurate. This is consistent with Connor-Desai and Reimers[37], who found that explaining the origins of misinformation as intentionally deceptive rather than unintentionally erroneous did not enhance correction effectiveness. Finally, when attributing novel claims to the sources post-correction in Experiment 2, we found that new false claims attributed to the doctor were still believed more than those attributed to the politician. This aligns with participants maintaining higher perceptions of credibility for doctors even after they were corrected.

Limitations and future directions

Several limitations of the current study must be noted. Although our decision to make the misinformation sources fictional had several advantages (e.g., participants did not hold strong priors about their credibility and false claims attributed to sources could be equated[38]), future work is required to examine how these effects extend to well-known sources who elicit strong prior beliefs (e.g., Dr. Anthony Fauci, J.D. Vance). Additionally, even though real-world assessments of doctors and politicians suggest that they differ substantially in perceived credibility[15], these two sources do not provide an unconfounded operationalization of "high" and "low" credibility sources. For example, doctors and politicians likely vary across dimensions unmeasured in this study (e.g., perceived bias[39]). Future research should extend our paradigm by including other high and low credibility sources to assess generalizability (e.g. scientists and teachers vs. social media influencers and advertising executives[15]). Furthermore, assessing credibility and belief across multiple timepoints (e.g., prior to misinformation exposure, after exposure but before correction) would be beneficial to better isolate the effects of corrections and priors about the sources. Regarding the generalizability of our results, we cannot be certain that the effects of corrections observed here would extend outside of the U.S. or to different cultural contexts. Even though doctors and politicians top and tail trustworthiness around the globe, future research should extend this work to countries beyond the U.S. to ensure replicability. Finally, while our samples were large, they may still have been underpowered for detecting higher-order interactions[40]. However, any undetected effects are likely to be small,and lacking practical relevance in the real world.

Practical recommendations

The findings of the current study have several practical implications. For one, our results suggest that corrections are effective for reducing misinformation belief and perceived credibility, regardless of source. Our findings also suggest that there is a reputational cost for politicians if they are wrong, particularly if they are found to be lying. Furthermore, if it is known that a source has *deliberately* spread falsehoods (i.e., disinformation sources), fact checkers and journalists should highlight the sources' intent to protect people from future disinformation. The finding that the doctor was perceived as relatively credible even after it was revealed that they were inaccurate also carries several practical implications for health communicators. Most notably, dubious health sources co-opting medical doctors' credibility may be penalized less when disseminating falsehoods. Finally, we show that conveying uncertainty can be damaging to credibility, and that hedging may not protect one from the consequences of being inaccurate. Medical professionals should be aware of this when communicating to the public and their patients.

Conclusion

In sum, our findings demonstrate that corrections consistently reduce belief in misinformation and perceived source credibility. After being shown to spread misinformation, high credibility sources may incur a larger credibility hit compared to low credibility sources. The results of Experiment 2 indicate that misinformation framing introduces additional complexity into how perceived source credibility is affected by corrections. These findings highlight the importance of considering why someone spreads misinformation in addition to who they are. Finally, expressing uncertainty reduced perceived credibility and offered no protective benefit against correction, illustrating the need for more effective strategies to convey doubt without necessarily compromising trust. Future research should consider the implications of these findings in the context of information ecosystems where low quality sources (e.g., influencers) can easily portray themselves as credible[41,42]. As the cues that distinguish high and low credibility sources become more blurry and trust in institutions wavers[43], it is increasingly important to consider the characteristics of the source alongside the veracity of their claims.

Data availability

All data necessary to replicate the results reported in this manuscript are available at https://osf.io/xthsg/.

Code availability

All code necessary to replicate the results and figures reported in this manuscript are available at https://osf.io/xthsg/.

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Author contributions

M.D., L.H.B., and B.S.T. designed the studies. M.D., L.H.B., and B.S.T recruited participants. M.D. conducted the

data analysis and visualized the data. M.D., L.H.B., and B.S.T. drafted the manuscript. L.H.B. and B.ST. provided critical revisions. All authors approved the final version of the manuscript for submission.

Declarations

Competing interests
The authors declare no competing interests.

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